

DONG-HO KANG

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EDUCATION

M.S. / Ph.D.	Sungkyunkwan University Electronic & Electrical Engineering Dissertation: Process Technology for 2D Semiconducting Devices (<i>Advisor: Prof. Jin-Hong Park</i>)	2019
B.S.	Sungkyunkwan University Electronic & Electrical Engineering	2014

APPOINTMENTS

Department Chair Gwangju Institute of Science & Technology Department of Semiconductor Engineering	Mar. 2026 – Present
Associate Professor Gwangju Institute of Science & Technology Department of Semiconductor Engineering	Mar. 2026 – Present
Assistant Professor Gwangju Institute of Science & Technology Department of Semiconductor Engineering	Mar. 2025 – Feb. 2026
Assistant Professor Gwangju Institute of Science & Technology School of Electrical Engineering and Computer Science	Nov. 2021 – Feb. 2025
Research Fellow Nanyang Technological University School of Electrical and Electronic Engineering	Mar. 2019 – Sep. 2021

RESEARCH INTEREST

Logic Devices — Process technologies for 2D materials — from synthesis and transfer to doping and junction control — toward next-generation logic devices.

Optoelectronic Devices — Neuromorphic and optoelectronic synaptic devices that emulate synapses and neurons for next-generation AI semiconductors.

System-level AI Hardware — In-sensor computing and spiking neural network hardware that realize AI semiconductors at the system level.

AWARDS

Young Scientist Award Korea Flexible & Printed Electronics Society(한국유연인쇄전자학회)	2025
Superior Research Award (Ph.D. Course), Second Prize Sungkyunkwan University <i>Electrical and Electronic Sector</i>	2017
Superior Research Award (M.S. Course), First Prize Sungkyunkwan University <i>Electrical and Electronic Sector</i>	2015
Superior Research Award (M.S. Course), First Prize Sungkyunkwan University <i>Electrical and Electronic Sector</i>	2014

SELECTED PUBLICATIONS

Top 10% Journal (JCR)

No.	Title	Journal	Authors
7	Revisiting Ferroelectric-Gated Phototransistors: A Tripartite Synapse-Inspired Approach to In-Sensor Image Processing	<i>Adv. Mater.</i> (2026)	Y. Lee, D. H. Seo ⁺ , J. S. Lee ⁺ , J. M. Jeon ⁺ , H. R. Kim, M. S. Kim, C. Ahn, S.-U. An, J. Choi, H. Kim, C. K. Jeong, H. Lin, <u>D.-H. Kang*</u> , Y. M. Song*
6	A Van der Waals Optoelectronic Synapse With Tunable Positive and Negative Post-Synaptic Current for Highly Accurate Spiking Neural Networks	<i>Adv. Funct. Mater.</i> (2026)	H. Yoon ⁺ , S. Park ⁺ , Y. K. Kim ⁺ , J. Baek, K. H. Kim, S. Yun, H. Son, J. Choi, B. C. Jang*, <u>D.-H. Kang*</u>
5	An Adaptive Inhibitory WSe2 Transistor for Retinomorphic In-Sensor Image Processing	<i>Adv. Funct. Mater.</i> (2026)	J. Baek ⁺ , J. Oh ⁺ , H. Kim, M. Park, H. Cho, J. Y. Park, H. Son, T. Kim, S.-U. An, Y. Zhang, H.-J. Shin, S. Oh*, <u>D.-H. Kang*</u>
4	Pseudo-magnetic field-induced slow carrier dynamics in periodically strained graphene	<i>Nat. Commun.</i> (2021)	<u>D.-H. Kang*</u> , H. Sun ⁺ , M. Luo ⁺ , K. Lu, M. Chen, Y. Kim, Y. Jung, X. Gao, S. J. Parluhan, J. Ge, S. W. Koh, D. Giovanni, T. C. Sum, Q. J. Wang, H. Li, D. Nam*, S.-H. Jo ⁺ , <u>D.-H. Kang*</u> , J. Shim, J. Jeon, M. H. Jeon, G. Yoo, J. Kim, J. Lee, G. Y. Yeom, S. Lee, H.-Y. Yu, C. Choi, J.-H. Park*
3	A High-Performance WSe2/h-BN Photodetector using a Triphenylphosphine (PPh3)-Based n-Doping Technique	<i>Adv. Mater.</i> (2016)	S.-H. Jo ⁺ , <u>D.-H. Kang*</u> , J. Shim, J. Jeon, M. H. Jeon, G. Yoo, J. Kim, J. Lee, G. Y. Yeom, S. Lee, H.-Y. Yu, C. Choi, J.-H. Park*
2	An ultrahigh-performance photodetector based on a perovskite--transition-metal-dichalcogenide hybrid structure	<i>Adv. Mater.</i> (2016)	<u>D.-H. Kang</u> , S. R. Pae, J. Shim, G. Yoo, J. Jeon, J. W. Leem, J. S. Yu, S. Lee, B. Shin*, J.-H. Park*
1	High-performance transition metal dichalcogenide photodetectors enhanced by self-assembled monolayer doping	<i>Adv. Funct. Mater.</i> (2015)	<u>D.-H. Kang*</u> , M.-S. Kim ⁺ , J. Shim, J. Jeon, H.-Y. Park, W.-S. Jung, H.-Y. Yu, C.-H. Pang, S. Lee, J.-H. Park*

FULL-PUBLICATION LIST (MAIN-AUTHORED)

Top 10% Journal (JCR)

No.	Title	Journal	Authors
25	Revisiting Ferroelectric-Gated Phototransistors: A Tripartite Synapse-Inspired Approach to In-Sensor Image Processing	<i>Adv. Mater.</i> (2026)	Y. Lee, D. H. Seo ⁺ , J. S. Lee ⁺ , J. M. Jeon ⁺ , H. R. Kim, M. S. Kim, C. Ahn, S.-U. An, J. Choi, H. Kim, C. K. Jeong, H. Lin, <u>D.-H. Kang*</u> , Y. M. Song*
24	A Van der Waals Optoelectronic Synapse With Tunable Positive and Negative Post-Synaptic Current for Highly Accurate Spiking Neural Networks	<i>Adv. Funct. Mater.</i> (2026)	H. Yoon ⁺ , S. Park ⁺ , Y. K. Kim ⁺ , J. Baek, K. H. Kim, S. Yun, H. Son, J. Choi, B. C. Jang*, <u>D.-H. Kang*</u>
23	(Invited Review) Spectrally Tunable 2D Material-Based Infrared Photodetectors for Intelligent Optoelectronics	<i>Adv. Funct. Mater.</i> (2026)	J. Ha ⁺ , Y. Ma ⁺ , Y. N. An ⁺ , S.-U. An ⁺ , H. H. Jung, S.-T. Varjamo, J. Yoo, J. Min, H. Kim, F. Ahmed, S. H. Chae, Y. M. Song, W. Cai, T. Hasan, Z. Sun, <u>D.-H. Kang*</u> , H.-J. Shin*, Y. Dai*, H. H. Yoon*
22	Avalanche-Enhanced Infrared Photodetection via Interlayer Absorption in a WSe2/MoS2 Heterostructure	<i>Nano Lett.</i> (2026)	B. Son ⁺ , H. Son ⁺ , Y. Kim, D. Kim, K. H. Kim, B. C. Jang, <u>D.-H. Kang*</u> , D. Nam*

No.	Title	Journal	Authors
21	An Adaptive Inhibitory WSe ₂ Transistor for RetinomorphiC In-Sensor Image Processing	<i>Adv. Funct. Mater.</i> (2026)	J. Baek ⁺ , J. Oh ⁺ , H. Kim, M. Park, H. Cho, J. Y. Park, H. Son, T. Kim, S.-U. An, Y. Zhang, H.-J. Shin, S. Oh*, D.-H. Kang*
20	Van der Waals hybrid-gate reconfigurable transistor integrating floating-gate logic and ferroelectric synaptic functions	<i>Small</i> (2026)	H. Son ⁺ , S. Lee ⁺ , S.-U. An, H. H. Jung, Y. N. An, G. Yu, D. Kwon, Y. Kim, D. Nam, H.-J. Shin, H. H. Yoon, I.-M. Yi, D.-H. Kang*
19	(Invited Review) 2D Materials in Logic Technology: Power Efficiency and Scalability in 2DM-MBC CFET	<i>Nano Lett.</i> (2025)	S. H. Shin ⁺ , D.-H. Kang* , H. H. Yoon ⁺ , J. Y. Park ⁺ , M. Song, H. Son, D. Ha*, H.-J. Shin*
18	(Invited Review) Enabling the Angstrom Era: 2D material-based multi-bridge-channel complementary field effect transistors	<i>npj 2D Mater. Appl.</i> (2025)	H. H. Yoon ⁺ , J. Y. Park ⁺ , Y. T. Megra ⁺ , J. H. Baek ⁺ , M. Song, D. Akinwande*, D. Ha*, D.-H. Kang* , H.-J. Shin*
17	Polarity Engineering in a Single MoTe ₂ Device for Homogeneous Complementary Circuit Applications	<i>ACS Appl. Mater. Interfaces</i> (2025)	H. Son, S. Lee, S. Lee, H. Go, J. Baek, D. Kim, S.-U. An, J. Lee, H. H. Yoon, H.-J. Shin, J.-H. Lee, H. Lim, I.-M. Yi, M. Kim*, D.-H. Kang*
16	Tactile neuromorphic system: convergence of triboelectric polymer sensor and ferroelectric polymer synapse	<i>ACS Nano</i> (2023)	H. Kim, S. Oh, H. Choo, D.-H. Kang* , J.-H. Park*
15	Pseudo-magnetic field-induced slow carrier dynamics in periodically strained graphene	<i>Nat. Commun.</i> (2021)	D.-H. Kang* , H. Sun ⁺ , M. Luo ⁺ , K. Lu, M. Chen, Y. Kim, Y. Jung, X. Gao, S. J. Parluhutan, J. Ge, S. W. Koh, D. Giovanni, T. C. Sum, Q. J. Wang, H. Li, D. Nam*
14	A neuromorphic device implemented on a salmon-DNA electrolyte and its application to artificial neural networks	<i>Adv. Sci.</i> (2019)	D.-H. Kang* , J.-H. Kim ⁺ , S. Oh, H.-Y. Park, S. R. Dugasani, B.-S. Kang, C. Choi, R. Choi, S. Lee, S. H. Park, K. Heo, J.-H. Park*
13	Rhenium Diselenide (ReSe ₂) Near-Infrared Photodetector: Performance Enhancement by Selective p-Doping Technique	<i>Adv. Sci.</i> (2019)	J. Kim ⁺ , K. Heo ⁺ , D.-H. Kang* , C. Shin, S. Lee, H.-Y. Yu, J.-H. Park*
12	Rhenium diselenide (ReSe ₂) infrared photodetector enhanced by (3-aminopropyl) trimethoxysilane (APTMS) treatment	<i>Org. Electron.</i> (2018)	M. H. Ali ⁺ , D.-H. Kang* , J.-H. Park*
11	Highly sensitive and reusable membraneless field-effect transistor (FET)-type tungsten diselenide (WSe ₂) biosensors	<i>ACS Appl. Mater. Interfaces</i> (2018)	H. W. Lee ⁺ , D.-H. Kang* , J. H. Cho, S. Lee, D.-H. Jun*, J.-H. Park*
10	(Invited Review) Recent progress in Van der Waals (vdW) heterojunction-based electronic and optoelectronic devices	<i>Carbon</i> (2018)	J. Shim ⁺ , D.-H. Kang* , Y. Kim, H. Kum, W. Kong, S.-H. Bae, I. Almansouri, K. Lee, J.-H. Park*, J. Kim*
9	Self-assembled layer (SAL)-based doping on black phosphorus (BP) transistor and photodetector	<i>ACS Photonics</i> (2017)	D.-H. Kang* , M. H. Jeon ⁺ , S. K. Jang, W.-Y. Choi, K. N. Kim, J. Kim, S. Lee, G. Y. Yeom, J.-H. Park*
8	Poly-4-vinylphenol (PVP) and poly (melamine-co-formaldehyde)(PMF)-based atomic switching device and its application to logic gate circuits with low operating voltage	<i>ACS Appl. Mater. Interfaces</i> (2017)	D.-H. Kang* , W.-Y. Choi ⁺ , H. Woo, S. Jang, H.-Y. Park, J. Shim, J.-W. Choi, S. Kim, S. Jeon, S. Lee, J.-H. Park*

No.	Title	Journal	Authors
7	Ultra-low doping on two-dimensional transition metal dichalcogenides using DNA nanostructure doped by a combination of lanthanide and metal ions	<i>Sci. Rep.</i> (2016)	D.-H. Kang , S. R. Dugasani, H.-Y. Park, J. Shim, B. Gnareddy, J. Jeon, S. Lee, Y. Roh, S. H. Park, J.-H. Park*
6	A High-Performance WSe ₂ /h-BN Photodetector using a Triphenylphosphine (PPh ₃)-Based n-Doping Technique	<i>Adv. Mater.</i> (2016)	S.-H. Jo ⁺ , D.-H. Kang ⁺ , J. Shim, J. Jeon, M. H. Jeon, G. Yoo, J. Kim, J. Lee, G. Y. Yeom, S. Lee, H.-Y. Yu, C. Choi, J.-H. Park*
5	An ultrahigh-performance photodetector based on a perovskite--transition-metal-dichalcogenide hybrid structure	<i>Adv. Mater.</i> (2016)	D.-H. Kang , S. R. Pae, J. Shim, G. Yoo, J. Jeon, J. W. Leem, J. S. Yu, S. Lee, B. Shin*, J.-H. Park*
4	Nondegenerate n-type doping phenomenon on molybdenum disulfide (MoS ₂) by zinc oxide (ZnO)	<i>Mater. Res. Bull.</i> (2016)	D.-H. Kang ⁺ , S.-T. Hong ⁺ , A. Oh, S.-H. Kim, H.-Y. Yu, J.-H. Park*
3	Controllable nondegenerate p-type doping of tungsten diselenide by octadecyltrichlorosilane	<i>ACS Nano</i> (2015)	D.-H. Kang , J. Shim, S. K. Jang, J. Jeon, M. H. Jeon, G. Y. Yeom, W.-S. Jung, Y. H. Jang, S. Lee, J.-H. Park*
2	High-performance transition metal dichalcogenide photodetectors enhanced by self-assembled monolayer doping	<i>Adv. Funct. Mater.</i> (2015)	D.-H. Kang ⁺ , M.-S. Kim ⁺ , J. Shim, J. Jeon, H.-Y. Park, W.-S. Jung, H.-Y. Yu, C.-H. Pang, S. Lee, J.-H. Park*
1	Indium (In)-and tin (Sn)-based metal induced crystallization (MIC) on amorphous germanium (α -Ge)	<i>Mater. Res. Bull.</i> (2014)	D.-H. Kang , J.-H. Park*

FULL-PUBLICATION LIST (CONTRIBUTING-AUTHORED)

Top 10% Journal (JCR)

No.	Title	Journal	Authors
27	Timing-Dependent Spiking Neural Network: Board-Level Hardware Implementation with Photoelectroactive Van der Waals Synapses	<i>Adv. Mater.</i> (2026)	S. Kim ⁺ , J.-I. Cho ⁺ , S. Lee ⁺ , Y. Shin, J.-J. Lee, T. Jang, H. Kim, J. Oh, S. Lee, K. Ko, J. Kang, J. Lee, M. T. Flavin, D.-H. Kang , B. C. Jang, J.-H. Ahn, Y. Lee, S. M. Won*, J.-H. Park*, S. Oh*
26	Configurable anti-ambipolar photoresponses for optoelectronic multi-valued logic gates	<i>Appl. Phys. Lett.</i> (2024)	X. Cui, S. Kim, F. Ahmed, M. Du, A. C. Liapis, J. A. Muñoz, A. M. Shafi, M. G. Uddin, F. Ali, Y. Zhang, D.-H. Kang , H. Lipsanen, S. Kang, H. H. Yoon*, Z. Sun* M. Luo ⁺ , H. Sun ⁺ , Z. Qi, K. Lu, M. Chen, D.-H. Kang , Y. Kim, D. Burt, X. Yu, C. Wang, Y. D. Kim, H. Wang, Q. J. Wang, D. Nam*
25	Triaxially strained suspended graphene for large-area pseudo-magnetic fields	<i>Optics letters</i> (2022)	D. Burt ⁺ , H.-J. Joo ⁺ , Y. Kim ⁺ , Y. Jung, M. Chen, M. Luo, D.-H. Kang , S. Assali, L. Zhang, B. Son, W. Fan, O. Moutanabbir, Z. Ikonik, C. S. Tan, Y.-C. Huang, D. Nam*
24	Direct bandgap GeSn nanowires enabled with ultrahigh tension from harnessing intrinsic compressive strain	<i>Appl. Phys. Lett.</i> (2022)	S. Seo, J. Koo, J.-W. Choi, K. Heo, M. Andreev, J.-J. Lee, J.-H. Lee, J.-I. Cho, H. Kim, G. Yoo, D.-H. Kang , J. Shim, J.-H. Park*
23	Controllable potential barrier for multiple negative-differential-transconductance and its application to multi-valued logic computing	<i>npj 2D Mater. Appl.</i> (2021)	

No.	Title	Journal	Authors
22	Biaxially strained germanium crossbeam with a high-quality optical cavity for on-chip laser applications	<i>Opt. Express</i> (2021)	Y. Jung ⁺ , Y. Kim ⁺ , D. Burt, H.-J. Joo, <u>D.-H. Kang</u> , M. Luo, M. Chen, L. Zhang, C. S. Tan, D. Nam*
21	Heterostrain-enabled dynamically tunable moiré superlattice in twisted bilayer graphene	<i>Sci. Rep.</i> (2021)	X. Gao ⁺ , H. Sun ⁺ , <u>D.-H. Kang</u> , C. Wang, Q. J. Wang, D. Nam*
20	1D photonic crystal direct bandgap GeSn-on-insulator laser	<i>Appl. Phys. Lett.</i> (2021)	H.-J. Joo ⁺ , Y. Kim ⁺ , D. Burt, Y. Jung, L. Zhang, M. Chen, S. J. Parluhutan, <u>D.-H. Kang</u> , C. Lee, S. Assali, Z. Ikonic, O. Moutanabbir, Y.-H. Cho, C. S. Tan, D. Nam*
19	Effect of large work function modulation of MoS ₂ by controllable chlorine doping using a remote plasma	<i>J. Mater. Chem. C</i> (2020)	K. H. Kim, K. S. Kim, Y. J. Ji, I. Moon, K. Heo, <u>D.-H. Kang</u> , K. N. Kim, W. J. Yoo, J.-H. Park, G. Y. Yeom*
18	Polarity control in a single transition metal dichalcogenide (TMD) transistor for homogeneous complementary logic circuits	<i>Nanoscale</i> (2019)	J. Shim ⁺ , S. woon Jang ⁺ , J.-H. Lim, H. Kim, <u>D.-H. Kang</u> , K.-H. Kim, S. Seo, K. Heo, C. Shin, H.-Y. Yu, S. Lee, D.-H. Ko, J.-H. Park*
17	Ultrasensitive MoS ₂ photodetector by serial nano-bridge multi-heterojunction	<i>Nat. Commun.</i> (2019)	K. S. Kim, Y. J. Ji, K. H. Kim, S. Choi, <u>D.-H. Kang</u> , K. Heo, S. Cho, S. Yim, S. Lee, J.-H. Park, Y. S. Jung, G. Y. Yeom*
16	Stable and reversible triphenylphosphine-based n-type doping technique for molybdenum disulfide (MoS ₂)	<i>ACS Appl. Mater. Interfaces</i> (2018)	K. Heo ⁺ , S.-H. Jo ⁺ , J. Shim, <u>D.-H. Kang</u> , J.-H. Kim, J.-H. Park*
15	Layer-controlled thinning of black phosphorus by an Ar ion beam	<i>J. Mater. Chem. C</i> (2017)	J. W. Park ⁺ , S. K. Jang ⁺ , <u>D. H. Kang</u> , D. San Kim, M. H. Jeon, W. O. Lee, K. S. Kim, S. J. Lee, J.-H. Park, K. N. Kim, G. Y. Yeom*
14	(Invited Review) Electronic and optoelectronic devices based on two-dimensional materials: From fabrication to application	<i>Adv. Electron. Mater.</i> (2017)	J. Shim, H.-Y. Park, <u>D.-H. Kang</u> , J.-O. Kim, S.-H. Jo, Y. Park, J.-H. Park*
13	Efficient threshold voltage adjustment technique by dielectric capping effect on MoS ₂ field-effect transistor	<i>IEEE Electron Device Lett.</i> (2017)	J. Park, <u>D.-H. Kang</u> , J.-K. Kim, J.-H. Park, H.-Y. Yu*
12	Extremely low contact resistance on graphene through n-Type doping and edge contact design	<i>Adv. Mater.</i> (2016)	H.-Y. Park, W.-S. Jung, <u>D.-H. Kang</u> , J. Jeon, G. Yoo, Y. Park, J. Lee, Y. H. Jang, J. Lee, S. Park, H. Yu, B. S. Shin, S. Lee, J. Park*
11	Extremely large gate modulation in vertical graphene/WSe ₂ heterojunction barristor based on a novel transport mechanism	<i>Adv. Mater.</i> (2016)	J. Shim, H. S. Kim, Y. S. Shim, <u>D.-H. Kang</u> , H.-Y. Park, J. Lee, J. Jeon, S. J. Jung, Y. J. Song, W.-S. Jung, J. Lee, S. Park, J. Kim, S. Lee, Y.-H. Kim, J.-H. Park*
10	Broad detection range rhenium diselenide photodetector enhanced by (3-aminopropyl) triethoxysilane and triphenylphosphine treatment	<i>Adv. Mater.</i> (2016)	S.-H. Jo, H.-Y. Park, <u>D.-H. Kang</u> , J. Shim, J. Jeon, S. Choi, M. Kim, Y. Park, J. Lee, Y. J. Song, S. Lee, J.-H. Park*

No.	Title	Journal	Authors
9	High-performance 2D rhenium disulfide (ReS ₂) transistors and photodetectors by oxygen plasma treatment	<i>Adv. Mater.</i> (2016)	J. Shim ⁺ , A. Oh ⁺ , D.-H. Kang , S. Oh, S. K. Jang, J. Jeon, M. H. Jeon, M. Kim, C. Choi, J. Lee, G. Y. Yeom, Y. J. Song, J.-H. Park*
8	Suppression of boron diffusion using carbon co-implantation in DRAM	<i>Mater. Res. Bull.</i> (2016)	S. H. Lee, S. G. Park, S. D. Kim, H.-C. Jung, I. G. Kim, D.-H. Kang , D. J. Kim, K. P. Lee, J. S. Choi, J.-W. Baek, M. Choi, Y. Park, C. Choi, J.-H. Park*
7	Improvement of the short channel effect in PMOSFETs using cold implantation	<i>Mater. Res. Bull.</i> (2016)	S. H. Lee, S. G. Park, S. H. Jeong, H.-C. Jung, I. G. Kim, D.-H. Kang , H.-J. Nam, D. J. Kim, K. P. Lee, J. S. Choi, W. Jung, Y. Park, C. Choi, J.-H. Park*
6	M-DNA/transition metal dichalcogenide hybrid structure-based bio-FET sensor with ultra-high sensitivity	<i>Sci. Rep.</i> (2016)	H.-Y. Park, S. R. Dugasani, D.-H. Kang , G. Yoo, J. Kim, B. Gnapareddy, J. Jeon, M. Kim, Y. J. Song, S. Lee, J. Heo, Y. J. Jeon, S. H. Park, J.-H. Park*
5	Phosphorene/rhenium disulfide heterojunction-based negative differential resistance device for multi-valued logic	<i>Nat. Commun.</i> (2016)	J. Shim, S. Oh, D.-H. Kang , S.-H. Jo, M. H. Ali, W.-Y. Choi, K. Heo, J. Jeon, S. Lee, M. Kim, Y. J. Song, J.-H. Park*
4	Wide-range controllable n-doping of molybdenum disulfide (MoS ₂) through thermal and optical activation	<i>ACS Nano</i> (2015)	H.-Y. Park, M.-H. Lim, J. Jeon, G. Yoo, D.-H. Kang , S. K. Jang, M. H. Jeon, Y. Lee, J. H. Cho, G. Y. Yeom, W.-S. J. Jung, J. Lee, S. Park, S. Lee, J.-H. Park*
3	Theoretical and Experimental Investigation of Graphene/High-k/p-Si Junctions	<i>IEEE Electron Device Lett.</i> (2015)	J. Shim, G. Yoo, D.-H. Kang , W.-S. Jung, Y.-C. Byun, H. Kim, W. T. Kang, W. J. Yu, H.-Y. Yu, Y. Park, J.-H. Park*
2	Germanium pin avalanche photodetector fabricated by point defect healing process	<i>Opt. Lett.</i> (2014)	J. Shim, D.-H. Kang , G. Yoo, S.-T. Hong, W.-S. Jung, B. J. Kuh, B. Lee, D. Shin, K. Ha, G. S. Kim, H.-Y. Yu, J. Baek, J.-H. Park*
1	n-and p-type doping phenomenon by artificial DNA and M-DNA on two-dimensional transition metal dichalcogenides	<i>ACS Nano</i> (2014)	H.-Y. Park, S. R. Dugasani, D.-H. Kang , J. Jeon, S. K. Jang, S. Lee, Y. Roh, S. H. Park, J.-H. Park*

PATENTS

No.	Title	Country	Inventors
1	부성 미분 전달컨덕턴스 특성을 갖는 반도체 소자 및 그 제조 방법	Korea (2019)	박진홍, 심재우, 강동호
2	Semiconductor Device with Negative Differential Transconductance and Method of Manufacturing the Same	USA (2020)	Jin-Hong Park, Jaewoo Shim, Dong-Ho Kang
3	인공 시냅스 소자 및 이의 제조방법	Korea (2021)	박진홍, 허근, 유광위, 구지완, 강동호
4	Bipolar Optical Synaptic Device	USA (2026)	Dong-Ho Kang , Hyejin Yoon, Soeun Park
5	양극성 광 시냅스 디바이스	Korea (2026)	강동호 , 윤혜진, 박소은

INVITED TALKS

No.	Title	Conference	Authors
8	(Invited) A Self-Inhibitory WSe ₂ Retinomorph Transistor for Adaptive In-Sensor Processing	<i>The 13th Korean Symposium on Graphene and 2D Materials</i> Busan, South Korea 2026.07.09 - 07.11	<u>D.-H. Kang*</u>
7	(Invited) Optoelectronic Synaptic Devices for in-Sensor Computing Applications	<i>The 22nd International Symposium on the Physics of Semiconductors and Applications (ISPSA 2026)</i> Jeju, South Korea 2026.06.29 - 07.02	<u>D.-H. Kang*</u>
6	(Invited) Van der Waals Heterostructure-Based Optoelectronic Devices for Neuromorphic and In-Sensor Applications	<i>한국전기전자재료학회 2026 하계학술대회</i> Busan, South Korea 2026.06.25 - 06.27	<u>D.-H. Kang*</u>
5	(Invited) Ferroelectric phototransistors for in-sensor processing toward all-day facial recognition	<i>10th IEEE Electron Devices Technology and Manufacturing (EDTM)</i> Penang, Malaysia 2026.03.01 - 03.04	<u>D.-H. Kang*</u>
4	(Invited) 인센서 컴퓨팅 응용을 위한 광-전자 시냅스 소자 연구	<i>2025 Photonic Conference</i> Pyeongchang, South Korea 2025.11.26 - 11.28	<u>D.-H. Kang*</u>
3	(Invited) TMD Photodetector: Performance Enhanced by Chemical Doping Technique	<i>14th Asia-Pacific Conference on Near-field Optics</i> Busan, South Korea 2023.06.19 - 06.22.	<u>D.-H. Kang*</u>
2	(Invited) Pseudo-magnetic fields in strained graphene for building optical quantum computing systems	<i>7th IEEE Electron Devices Technology and Manufacturing (EDTM)</i> Seoul, South Korea 2023.03.07 - 03.10.	<u>D.-H. Kang*</u>
1	(Invited) A Study of Carrier Dynamics in Pseudo-Landau-Quantized Graphene	<i>한국광학회 2023 동계학술대회</i> Busan, South Korea 2023.02.14 - 02.17.	<u>D.-H. Kang*</u>